

First Flight — a tester's guide to version 1.0

What the instrument measures, what every screen is telling you, and a set of structured experiments — real-time checks, breathing and meditation comparisons, exercise sessions — to put the numbers under honest pressure.

VERSION 1.0 · TESTFLIGHT AUGUST 2026 REQUIRES POLAR H10 CHEST STRAP

01 · WHAT WYTHIN IS

A nervous-system instrument, not a wellness app

Wythin pairs with a **Polar H10 chest strap** and takes the raw feed — ECG at 130 Hz, chest motion at 200 Hz, every beat-to-beat interval. From that stream it computes a full autonomic profile **every two seconds**, all day (every 30 seconds when the app is in the background), and translates the physiology into plain language: your capacity this morning, your state right now, the true cost of a workout, and whether ten minutes of slow breathing actually landed.

The key capabilities you are testing:

CAPABILITY	WHAT IT DOES
Live metrics	Nine core measures (plus ~10 supporting ones) recomputed every 2 s: Vagal Tone (DC), Energy Reserve (HRV/RMSSD), Inner Noise (PIP), Harmony (DFA α1), Conscious Breathing (RSA), Stress Balance, Adaptive Capacity (RCMSE), Calm Power (VTI), Pulse.
Live state	One of nine named states ("Locked In", "Calm & Alert", "Running On Empty"...), classified against <i>your own</i> baseline on three axes — energy, tension, recovery — with a WHY panel and one suggested action.
Morning anchor	The day's first genuinely rested window (≥3 quiet minutes after 04:00) becomes a frozen daily baseline, and produces Today's Potential (0–100, capped at 75 in ambiguous deep-rest physiology).
Guided practice	Three pacers — Box 6-6-6-6, Resonance 6/min, Coherent ~5.5/min — with metronome, adjustable length/pace, and study citations.
Session scoring	Log anything (8 types, 60+ subtypes). Each session captures 5 min <i>before</i> , the session <i>during</i> , 10 min <i>after</i> — for all nine metrics. Restorative sessions get an impact % and 0–100 practice score; exercise gets the five-question read and a session score built from your own history.

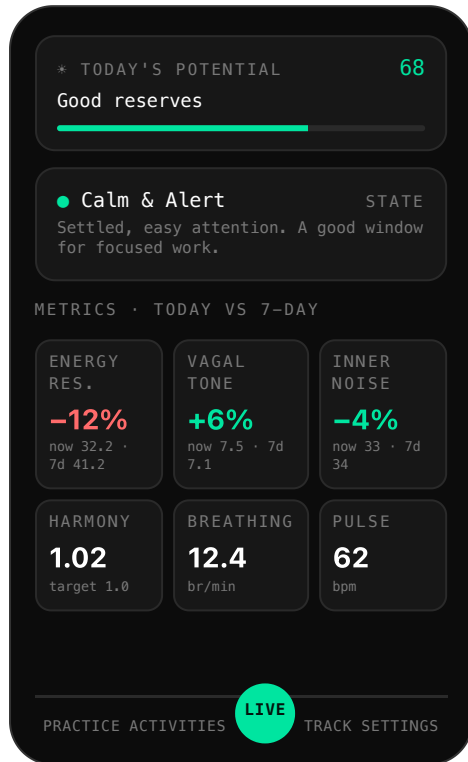
Trends	Track tab: week / month / 6-month bars per metric against your own 90-day baseline, plus consistency (practice minutes, wear hours, streak).
Nudges	At most 3/day, never at night, never while moving — e.g. "You've been running hot" with a 5-minute intervention.
Your data	Private sync, personal access tokens, an MCP server so you can query your own physiology from Claude Code, and one-tap erase of everything.

WHAT YOU NEED BEFORE DAY ONE

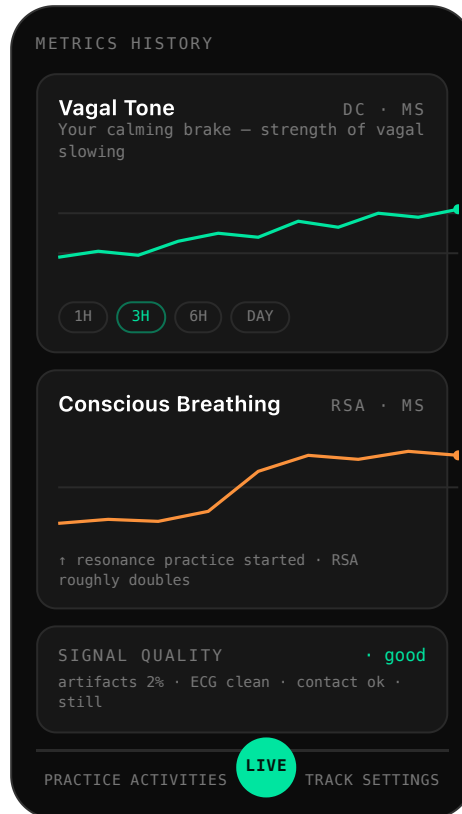
- Polar H10, worn snug just below the chest muscles; **wet the electrode strip** before each wear — dry electrodes are the #1 source of artifacts.
- A fresh-ish CR2025 coin cell. The app warns below 5% — the strap can die mid-session.
- Bluetooth permission granted; keep the phone within a few metres for the live feed.
- Plan to wear the strap for a few longer stretches (a morning, a workout + 10 min after, one ordinary afternoon) — the app's comparisons need at least 2 reference days with ≥ 30 min of wear.

Five tabs, and what each screen is telling you

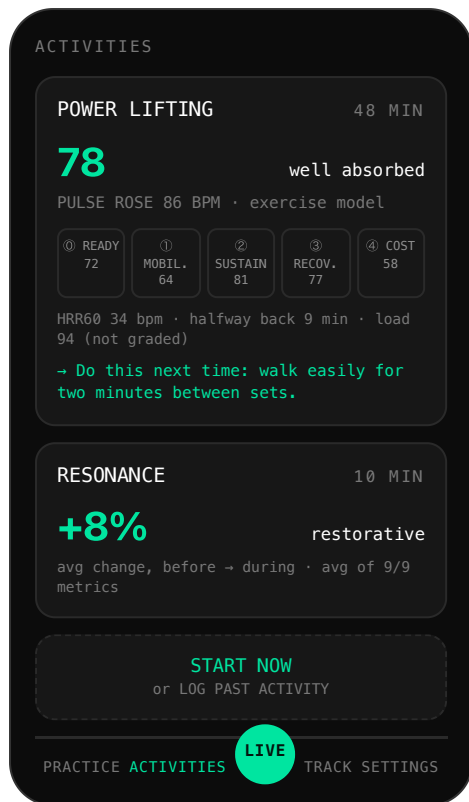
Five tabs: **Practice** · **Activities** · **Live** · **Track** · **Settings**, with Live as the prominent centre button. The mock-ups below are drawn from the codebase to orient you — they are illustrative, not pixel-perfect screenshots.



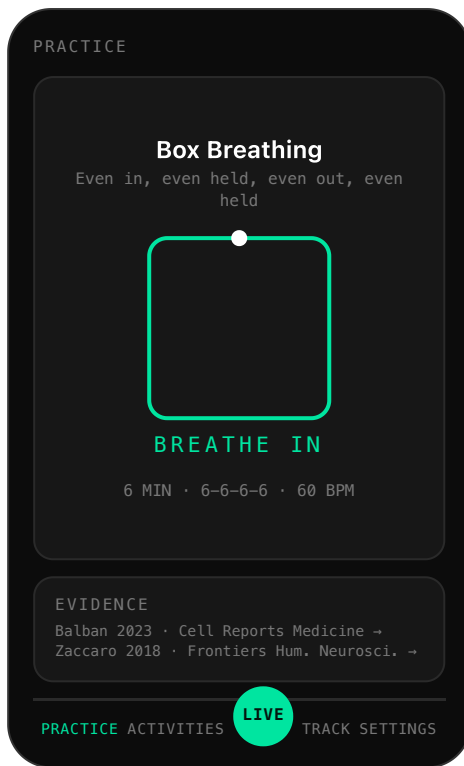
Live tab. Potential strip, current state, and the nine-tile grid — hero number is today vs your last 7 recorded days; the footnote carries the absolutes.



Metrics history. ~19 chart cards with reference bands, a shared time-window control, and the signal-quality panel that tells you when to trust a reading.



Activities. Exercise rows show the session score and the five-question read; restorative rows show impact % with its denominator. Note the one concrete instruction at the end.



Practice. The animated pacer (square for four phases, ring for two) with a metronome cue, and evidence cards linking the actual studies.

Seven experiments for the live feed

These are short, repeatable probes of the 2-second pipeline. For each: note the time, do the thing, watch the Live tab. The metrics have different warm-up times by design — that ladder is itself the first test.

TEST 1 The warm-up ladder

DO

Strap on, sit quietly, start a timer.
Watch tiles populate.

EXPECT

Pulse & HRV almost immediately ·
Inner Noise ~30 s · Adaptive
Capacity ~1½ min · Vagal Tone
~2½ min of clean signal.

VERIFY

Tiles that aren't ready say so
honestly rather than showing junk.
Note any metric that never fills
while signal quality reads good.

TEST 2 The resonance probe — the app's central claim

DO

5 min normal breathing, then start
the Resonance pacer (6/min) for
5–10 min, eyes on the Live charts.

EXPECT

Conscious Breathing (RSA) climbs
— often 2–3× within minutes.
Coherence heads past 0.6. Breath
rate reads ~6/min.

VERIFY

**Stress Balance should NOT
spike.** It was deliberately built so
slow breathing doesn't read as
tension. If it climbs into the 65+
zone during calm paced breathing,
that's a bug worth reporting.

TEST 3 Talking vs. silence

DO

2 min sitting silently, then 2 min
reading aloud or on a call, then
silence again.

EXPECT

Speech makes breathing irregular:
RSA drops, breath-rate chart gets
ragged, coherence falls.
Everything recovers within a
couple of minutes of silence.

VERIFY

The state row shouldn't lurch to an
alarming state from one phone call
— hysteresis should hold it steady.

TEST 4 Stand up (orthostatic probe)

DO

3 min seated, then stand still for
2 min.

EXPECT

Pulse jumps ~10–20 bpm within
seconds, Energy Reserve dips,
then both partially settle while you
stay standing.

VERIFY

The 2-second cadence: the pulse
tile should visibly move within a
tick or two of standing.

TEST 5 The espresso

DO

Log Drinks → Espresso with
START NOW; keep the strap on for
45 min.

EXPECT

Pulse drifts up a few bpm over 20–
40 min; Energy Reserve and Vagal
Tone often ease down; Inner Noise
may rise.

VERIFY

The session impact should land in
the "steady" or "stirred" range —
an honest reading for a stimulant.
Repeat on a different day: is the
response consistent for you?

TEST 6 Motion and bad signal — try to fool it

DO

Walk briskly for 2 min; rub the
strap; let one electrode go half-
dry; briefly walk out of Bluetooth
range.

EXPECT

The Signal Quality panel should
call it out (artifact %, contact,
motion), and scores that need
stillness should decline to
compute rather than degrade
silently.

VERIFY

Reconnection after signal loss:
does live data resume cleanly?
Does a running activity mark the
gap instead of inventing data?

TEST 7 Cold on the face (dive reflex)

DO

Splash genuinely cold water on
your face, or hold a cold pack to
cheeks/forehead ~30 s while
seated. (Skip if you have any
cardiac condition.)

EXPECT

A brief, visible heart-rate dip and a
vagal surge — one of the few ways
to move Vagal Tone on demand.

VERIFY

Do the charts catch a ~30-second
event at the 2-second resolution?

Testing the day's most important five minutes

The anchor is the standardised condition every daily comparison is built on. Protocol: **strap on within ~15 minutes of waking (after 04:00), then 3–5 minutes genuinely still** — sitting or lying, no phone scrolling, unforced breathing. The stillness gate is strict (accelerometer variance under 20 mg), and gaps over 2 minutes split the window.

- ☐ Does an anchor form on a normal morning? (Potential strip appears with a score and confidence.)
- ☐ Is it **frozen**? A calmer sit at 11:00 should *not* improve the number.
- ☐ Slow-breathing mornings: resonance breathing at ~6/min during the anchor window should still be accepted (the gate was widened to 4–20 br/min for exactly this).
- ☐ A late anchor (after noon) should be marked late / lower confidence, and an anchor at a very unusual hour should read "not comparable" rather than pretending.
- ☐ Fidget mid-window: more than ~45 s of stirring should push the anchor to a later window.
- ☐ Skip one morning in a week — the streak should survive (one missed day per rolling 7 is forgiven). Skip two — it shouldn't.
- ☐ After ~8 mornings, watch for the stability penalty becoming active: does an erratic week read lower than a steady one at the same average?

THE CAP IS A FEATURE

If you ever see the Potential score stop at **75** on a morning of unusually low resting heart rate, that's the deep-rest guard working — the app caps rather than celebrates physiology it can't distinguish from over-recovery. Please note the morning it happens.

Blind-rate first, then look

The most valuable observations come from rating yourself *before* looking at the app. Each morning after your anchor (before opening the strip), and once mid-afternoon, jot three 0–10 ratings — then open the app and record what it says:

YOUR BLIND RATING	COMPARE AGAINST	WHAT A CORRELATION WOULD LOOK LIKE
Energy (0–10)	Today's Potential; Energy Reserve tile	Your 8-energy days cluster above your 4-energy days over ~2 weeks. Perfect tracking is not expected — note the misses especially.
Tension (0–10)	Stress Balance; Inner Noise	Deadline days sit visibly higher; wind-down evenings lower.
Focus quality (0–10)	Live state; Harmony (DFA α1)	Do your subjectively "locked in" work blocks coincide with the engaged states and Harmony near 1.0?

Then annotate the obvious life events and watch the next morning's anchor:

- **Alcohol in the evening** — the classic: expect a lower Energy Reserve / Vagal Tone anchor and a lower Potential next morning. Dose-dependent?

- **Short night** (under ~6 h) — does Potential drop even when you subjectively feel "fine"? That divergence is exactly what the instrument is for.
- **Hard training day** — next morning dip, second-morning rebound. Does the two-day shape appear?
- **Late heavy meal, stressful conversation, long screen evening** — smaller effects; note whether anything shows at all.
- **A genuinely good day** — does "Renewed / Thriving" ever fire, and did it feel earned?

Log any day where the app's read and your felt state *strongly disagree* — date, time, what it said, what you felt. Those are either bugs, calibration gaps, or the app noticing something before you do; all three are worth knowing.

Questions the impact score should be able to answer

Every logged practice produces an impact % (before → during), a practice score (which also counts whether gains *held* in the 10 minutes after), and the per-metric breakdown. Keep the strap on for 10 minutes after every session — the "after" window is where the interesting scoring lives.

1. **Pacer bake-off.** On alternating days at the same hour: Box (6 min) vs Resonance (10 min) vs Coherent (12 min), three rounds each. Which wins on impact %? Which wins on the *after* phase — whose calm survives standing up?
2. **Find your resonance rate.** Run Resonance at 6/min, then Coherent at ~5.5/min. Which produces higher Conscious Breathing (RSA) and Coherence *for you*? The difference can be real and personal.
3. **Meditation styles.** Log Vipassana vs Body Scan vs Guided (same length, same time of day). Does one consistently move Inner Noise or Vagal Tone more? Does your subjectively "deepest" sit match the highest score?
4. **Dose curve.** 5 vs 10 vs 20 minutes of the same practice across a week. Does impact grow with length, plateau, or peak early? Where's your minimum effective dose?
5. **Time of day.** The identical practice at 08:00 vs 14:00 vs 21:00. Same impact? (Before-windows differ wildly — this also stress-tests the before/during comparison logic.)
6. **The honesty check.** Do one deliberately bad session — fidget, check your phone, breathe irregularly. It should score near "held steady", not "restorative". An app that flatters a bad sit can't be trusted on a good one.
7. **Wim Hof / fast breathwork (if it's your practice).** Expect the *opposite* signature during: pulse up, HRV down — it's a stressor by design. Does the pulse-rise rule (≥ 15 bpm) route it to the exercise model? Does the after-window show the rebound?

Putting the five questions under load

Exercise sessions are scored as Recovery 45% · Cost 35% · Efficiency 20%, each a percentile against your own history in that same subtype (minimum 3 sessions before scores appear — the first two are calibration). Load is shown but never graded.

1. **The classification rule.** Log a brisk vinyasa as "Yoga" and a genuinely restorative stretch as "Exercise". The first (pulse +15 bpm or more) should get the exercise analysis; the second should not. The label must lose to the pulse both ways.
2. **Same route, three weeks.** Repeat one easy run weekly. Watch the recovery half-time and HRR60 across repetitions — the percentile scores should start meaning something at session 3.
3. **Easy vs intervals.** Compare the five-question reads: intervals should cost more (Ⓔ), recover slower (Ⓕ) — but on a well-recovered day, not catastrophically. Does the read match how the sessions felt?
4. **Readiness → cost.** On a morning the app says you "arrived under-recovered", train anyway (sensibly). Does the session read costlier than the same workout on a good-readiness day? That's the core claim of the readiness index.

5. **The 10-minute tail.** Stay strapped and still for 10 min after finishing. Published comparators the app is built against: ~48% vagal reactivation after yoga, ~28% after resistance, ~21% after aerobic work. Does your pattern rank the same way across modalities?
6. **Thermal contrast.** Sauna (activating — pulse rises!) vs cold shower (usually restorative-classified). Check both the classification and the after-effects on Vagal Tone.
7. **Zones sanity.** During a hard effort, does the lock-screen Live Activity zone (1–5) match perceived effort? Does the grey-zone warning appear on those neither-easy-nor-hard runs?

SAFETY NOTE

Nothing here asks you to train harder than you would anyway. The under-recovered-day test means your normal session, not a maximal one; skip the cold/breath-hold tests if you have any cardiovascular condition. Wythin is a wellness instrument, not a medical device.

Edges, plumbing, and the data promise

SCENARIO	WHAT TO CHECK
Background wear	Phone locked in your pocket for 2 h: history should fill at the 30-second background cadence with no big gaps; battery drain should be tolerable. Note both.
Live Activity	Start a session, lock the phone: heart rate, % HR-reserve and zone on the lock screen / Dynamic Island; a clear "strap lost" indication when you walk away from the phone.
Mid-session strap loss	Take the strap off for 3 min inside a logged session. Gaps over ~90 s must contribute nothing to load — the session should survive with an honest hole, not fabricated beats.
Still-recording card	Forget to stop an activity; verify the "STILL RECORDING" card surfaces it and ending it retroactively produces sane windows.
Nudges	Over a working week: do "running hot" nudges fire at sensible moments? Never more than 3/day, never 22:00–07:00, never mid-walk. Do the suggested 5-minute interventions match the trigger?
Battery warning	If your CR2025 runs low: the sub-5% alert should arrive before the strap dies mid-session.
Sync & continuity	Force-quit and relaunch mid-day: history intact? Delete and reinstall (after a sync): does your server history survive the phone?
Claude Code access	Settings → create a token, run the <code>claude mcp add</code> command it shows you, then ask Claude: <i>"how did my vagal tone respond to sauna this month?"</i> Verify the numbers match the app — and that a revoked token stops working.
The erase promise	(End of test period, if you're willing) Delete all data from Settings; confirm the API returns nothing afterwards.

What a useful report looks like

Because every 2-second sample is synced, a timestamp is worth a thousand words — the raw window around any oddity can be pulled from the server.

- **Date + local time** of the observation (minute precision is enough).
- **Where:** which tab / card / metric.
- **Expected vs seen** — one line each. Screenshots help but aren't required.
- **Context:** strap freshly wetted? moving? phone in background? battery level?
- For scoring disagreements: what the app said, what you felt, and which experiment above you were running.

Equally valuable as bugs: the correlation journal from section 05, your pacer bake-off results, and any moment the app told you something about yourself before you knew it.

is a wellness instrument, not a medical device; nothing in this guide is medical advice. Expected physiological responses are typical patterns – individual responses vary, and documenting your variation is the point of testing.